Experiment 5: Vacuum Cleaner Problem

Aim

To implement and solve the Vacuum Cleaner Problem using Artificial Intelligence techniques.

Algorithm

1. Start
2. Initialize the problem state
3. Apply valid operations/search
4. Repeat until goal state is reached
5. Display solution
6. Stop

Flowchart

```
Start
  |
Initialize
  |
Process/Search
  |
Goal Reached?
 /  \
Yes  No
 |    |
Print Repeat
 |
Stop
```

Objective

- Understand state-space search.
- Implement AI search technique.
- Obtain the required goal state.

Program

```
rooms = {}



rooms["A"] = "Dirty"

rooms["B"] = "Dirty"



for room in rooms:
```

```
    if rooms[room] == "Dirty":
        print("Cleaning", room)
        rooms[room] = "Clean"

print(rooms)
```

Output

Result

The program was executed successfully.

Conclusion

The program demonstrated the AI concept successfully and produced the expected result.